A decorative graphic on the left side of the slide, consisting of a black crosshair overlaid on a blue, red, and yellow gradient background.

Computer Programming II (CSE2035)

Week 5 Lab Session:

Object Oriented Programming Part I

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Homework 5-1: My String

- Let's write a string class that works like the C++ string
- Properties, constructor, and Print() are given in my-string.hpp file (**do not change this file**)

```
class MyString {
private:
    char *arr;
    size_t length;
public:
    MyString(const char *s) {
        length = strlen(s);
        if (length == 0) { arr = NULL; }
        else {
            arr = (char*) malloc(length);
            memcpy(arr, s, length); // No '\0' needed here!
        }
    }
}
```

Homework 5-1: My String

- **Implement the following functions in my-string.cpp**
 - If an invalid 'idx' or 'n' argument is given in Insert() or Remove(), don't do anything and just return.
 - Make sure that there is no buffer overflow in your code

```
// Should not leave any heap-allocated memory.
~MyString();
// Get the length of this string.
size_t GetLength();
// Make this string into an empty string.
void Clear();
// Insert 's' at position 'idx' of this string.
void Insert(MyString *s, int idx);
// Starting from the position 'idx', remove 'n' characters.
void Remove(int idx, int n);
```

Homework 5-1: My String

- **Make sure that there is no memory leak**
 - After test() returns, there shouldn't be any heap-allocated memory that is not free()-ed yet.

```
void test() {
    MyString s1("ABC");
    MyString s2(" DEF");
    s1.Insert(&s2, 3);
    s1.Print(); // Should print "ABC DEF".
    cout << s1.GetLength() << endl; // Should print "7".
}

int main() {
    test();
    return 0;
}
```

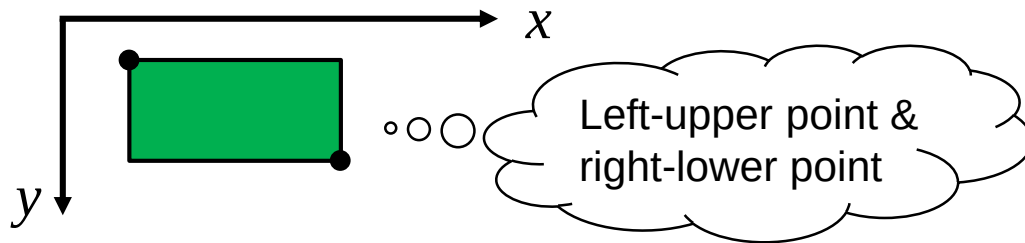
Homework 5-2: Simple Graphics

- Let's write a simple graphics class for drawing figures
- Point class is already defined in `graphics.hpp` (**do not change this file**)
 - A point is represented with an x-y coordinate. Note that in computer systems, (0, 0) usually means top-left position (not bottom-left).
 - Property 'fixed' represents whether this point is movable or not.

```
class Point {
private:
    int x;
    int y;
    bool fixed;
public:
    ...
}
```

Homework 5-2: Simple Graphics

- **Rectangle class is defined by using two Point objects**
 - Properties, constructor, and Print() are given in graphics.hpp

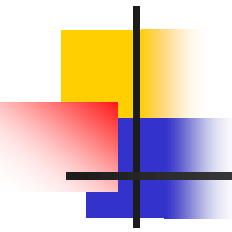


```
class Rectangle {  
private:  
    Point left_up;  
    Point right_low;  
public:  
    // You can use 'member initializer' to call constructor of 'Point'.  
    Rectangle(int left_up_x, int left_up_y,  
              int right_low_x, int right_low_y)  
        : left_up(left_up_x, left_up_y),  
          right_low(right_low_x, right_low_y)  
    { }
```

Homework 5-2: Simple Graphics

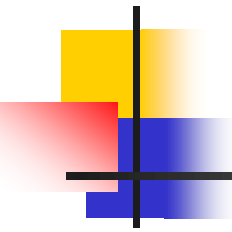
- Implement the following functions of Rectangle class in `graphics.cpp`
 - If a negative 'delta' value is given in `IncreaseHeight()` or `IncreaseWidth()`, don't do anything and just return.

```
// Make the rectangle unmovable/unresizable.
void Fix();
// Make the rectangle movable/resizable.
void Unfix();
// Increase the height by 'delta'. Left-upper point should not change.
void IncreaseHeight(int delta);
// Increase the width by 'delta'. Left-upper point should not change.
void IncreaseWidth(int delta);
// Compute the size (area) of the rectangle.
int Size();
```



Submission Guideline

- **Submit the completed `my-string.cpp` and `graphics.cpp` to the “Week 5 Lab Assignment” page in cyber campus**
 - Remember that you are allowed to change only these files
 - Do not zip the directory, just upload this file in the form
 - If your submission format is wrong, 10% deduction
 - If the submitted file doesn't compile with the "make" command, cannot give you any point
- **Deadline is 10/5 (Wednesday) 23:59**
- **Late submission is allowed until 10/7 (Friday) 23:59**
 - 20% deduction

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Q & A Session
